NATIONAL MATHEMATICS CONTEST
A-LEVEL PAPER 1 QUESTIONS AND SOLUTIONS.

SATURDAY 5TH APRIL, 2025 9:00 AM - 12:15 PM

SECTION A: 5 marks each

1. What is the value of $\sum_{k=1}^{2025} \log\left(1-\frac{1}{k+1}\right)$?

   Solution : $\log\left(1-\frac{1}{k+1}\right) = \log\left(\frac{k}{k+1}\right) = \log k - \log(k+1)$, *thus* $\sum_{k=1}^{2025} \log\left(1-\frac{k}{k+1}\right) = \sum_{k=1}^{2025} (\log k - \log(k+1))$ *which will cancel each other when you open (telescoping sum) and the only surviving term is* $-\log(2026)$. ■

2. Prove that: $\frac{yx+z^2}{xz} \geqslant 3-\frac{x}{y}$, for all real numbers $x, y, z > 0$.

   Solution : *Since* $x, y, z > 0$, *it follows that* $\frac{x}{y}+\frac{y}{z}+\frac{z}{x} \geqslant 3\sqrt[3]{\frac{x}{y}\cdot\frac{y}{z}\cdot\frac{z}{x}} = 3$ *by the A.M-G.M inequality. It then follows that* $\frac{yx+z^2}{xz} = \frac{y}{z}+\frac{z}{x} = \frac{x}{y}+\frac{y}{x}+\frac{z}{x}-\frac{x}{y} \geqslant 3-\frac{x}{y}$, *as required.* ■

3. A triangle has sides of length at most 2, 3 and 4 respectively. Determine, with proof, the maximum possible area of the triangle.

   Solution : *Let* θ *be the largest angle of the possible triangle. Therefore,* θ *is the angle between the two shorter sides, of lengths* a *and* b. *The area of the triangle is given by* $\text{Area} = \frac{1}{2}ab\sin(\theta)$, *where* $a \leqslant b$ *are the shorter sides (as stated before). However, we have* $a \leqslant 2$, $b \leqslant 3$, *and* $\frac{\sqrt{15}}{4} \leqslant \sin(\theta) \leqslant 1$ *as* $\cos(\theta) = \frac{2^2+3^2-c^2}{2\cdot 2\cdot 3} \geqslant \frac{2^2+3^2-4^2}{12} = \frac{-1}{4}$. *Therefore,* $\text{Area} \leqslant 3$, *with equality when the triangle is right angled with legs* 2 *and* 3. *This gives a hypotenuse of length* $\sqrt{13} < 4$, *which is valid. Therefore, the maximum area is 3.* ■

4. Given that

   $$34! = 295232799039a041408476186096435b0000000,$$

   in decimal representation, find the numbers a and b.

   Solution :

   - *We know that* $9|34!$. *Also we know from the rule of divisibility by* 9 *that* $9|n$ *if and only if* $9|S(n)$, *where* $S(n)$ *is the sum of digits of* n. *Here,* $S(34!) = a+b+136$. *Hence,* $9 \mid a+b+136$ *and so,* $a+b \in \{8, 17\}$ *as* $0 \leq a, b \leq 9$.
   - *Similarly,* $11|34!$ *and the rule of divisibility of* 11 *we know that* $11|n$ *if and only if* $11|T(n)$, *where* $T(n)$ *is the alternating sum of the digits of* n. *Here,* $T(34!) = a-b+18$ *Therefore, we have* $11 \mid a-b+18$ *and so,* $a-b \in \{-7, 4\}$ *as* $0 \leq a, b \leq 9$.
   - *Now, note that* $a+b$ *and* $a-b$ *have the same parity. Hence, the only possibilities are* $(a+b, a-b) = (8, 4)$ *and* $(17, -7)$, *and the two cases yield* $(a, b) = (6, 2)$ *or* $(a, b) = (5, 12)$, *respectively. But since* $a, b \leq 9$, *hence the second case is impossible. Thus,* $\boxed{(a, b) = (6, 2)}$, *proving the claim.*

   ■

5. John has a slow bicycle tire leak. He hasn't bothered to fix it because it takes the tire 9 days to go flat after filling it. John acquired a second puncture in the same tire, and now the tire with both punctures will flatten in 3.6 days. What is the time in days it takes the second puncture alone to flatten the tire?

   Solution : *The rate of leaking with one puncture that take* 9 *days to be flat is* $\frac{1}{9}$ *and the rate of leaking with the second puncture that take* x *days to be flat is* $\frac{1}{x}$ *and the combined rate is* $\frac{1}{3.6}$; *now* $\left(\frac{1}{3.6} = \frac{1}{9}+\frac{1}{x}\right)$ *solving for* x *gives* $x = 6$ *days.* ■

6. An N-meanset is any set of distinct positive integers including N, while α_N is the least possible mean of any N-set. For how many values of $1 \leqslant n \leqslant 2025$ is α_n an integer?

   Solution : *If* $\alpha_N = m$ *is an integer, then we may add* m *if it is missing, or remove* m *if it is present,*

without changing the mean. Moreover if m is minimal, then the N-set can not contain an integer between m and N since removing it would reduce the mean. On the other hand, it must contain every integer below m since adding in any such integer would reduce the mean. It follows that $\frac{1+2+\cdot+m-1+N}{m} = \frac{1+2+\cdot+m+N}{m+1} = m$. *This rearranges to* $N = \frac{m(m+1)}{2}$, *so* α_N *is an integer if and only if* N *is a triangle number. Since* $N \leqslant 2025$, *it follows that* $2050 \geqslant N = \frac{m(m+1)}{2} > \frac{m^2}{2}$, *hence* $-63 \leqslant m \leqslant 63$. *Since* $1 \leqslant m \leqslant 63$, *it follows that there are 63 triangle numbers below 2025.* ■

7. An obtuse angled triangle ABC has side lengths a, b, and c opposite $\angle$A, $\angle$B, and $\angle$C, respectively. Show that $a^3\cos(A) + b^3\cos(B) + c^3\cos(C) < abc$.

   Solution : *If* $\angle C$ *is obtuse, then by the cosine rule,* $-1 < \cos(C) < 0$ *hence* $a^2 + b^2 - c^2 < 0$. *Also* $a^3\cos(A) + b^3\cos(B) + c^3\cos(C) = a^3 \cdot \frac{b^2+c^2-a^2}{2bc} + b^3 \cdot \frac{a^2+c^2-b^2}{2ac} + c^3 \cdot \frac{a^2+b^2-c^2}{2ab} = \frac{a^4(b^2+c^2-a^2)+b^4(a^2+c^2-b^2)+c^4(a^2+b^2-c^2)}{2abc} = \frac{(a^2+b^2-c^2)(a^2+c^2-b^2)(b^2+c^2-a^2)+2a^2b^2c^2}{2abc} < \frac{2a^2b^2c^2}{2abc} = abc.$ ■

8. Find all pairs of integers (x, y) such that $x^3 - y^3 = xy + 61$.

   Solution : *Using the fact that:*

   $$x^3 - y^3 = (x-y)(x^2+xy+y^2) = xy + 61.$$

   Notice that $x > y$. *Therefore, we have to consider* $x^2 + xy + y^2 \leq xy+61$, *or* $x^2+y^2 \leq 61$. *Since* $x > y$, *we have:*

   $$61 \geq x^2 + y^2 \geq 2y^2 \quad \Rightarrow y^2 \leq 31.5 \quad \Rightarrow y \in \{1, 2, 3, 4, 5\}.$$

   Thus, we have the following possible equations to solve:

   $$\begin{aligned} y = 1, &\quad x^3 - x - 62 = 0, \\ y = 2, &\quad x^3 - 2x - 69 = 0, \\ y = 3, &\quad x^3 - 3x - 88 = 0, \\ y = 4, &\quad x^3 - 4x - 125 = 0, \\ y = 5, &\quad x^3 - 5x - 186 = 0. \end{aligned}$$

   Of the above equations, we see that the only working integer value for x *is when* $x = 6$ *and* $y = 5$. *So, the only possible pair of solutions is* $(x, y) = (6, 5)$. ■

9. In a recently completed Fistula awareness run, a total of n participants competed in a 5-kilometer marathon held in Kampala, Uganda. The participants were organized into k teams, with each team consisting of exactly 4 participants. For each team, one participant was selected as the team captain. Given that the event organizer had to assign participants to these teams and select captains, how many ways could this assignment have been made?

   Solution 1 : *The total number of participants in the marathon is* n, *and they are divided into* k *teams, each consisting of exactly 4 participants. This implies that* $4k$. *To determine the number of ways to assign participants into these teams, we first count the number of ways to partition* n *participants into* k *teams of 4. This is given by the formula* $\frac{n!}{(4!)^k k!}$ *where* $n!$ *accounts for arranging all participants,* $(4!)^k$ *corrects for the ordering within each team (since the order of members in a team does not matter), and* $k!$ *accounts for the indistinguishability of the teams themselves. Once the teams are formed, a captain must be selected from each team. Since each team consists of 4 participants, there are 4 ways to choose a captain for each team, leading to a total of* 4^k *ways to assign captains across all teams. Since the team formation and captain selection are independent choices, we multiply these values, yielding the final expression:* $\frac{(4k)!4^k}{(4!)^k k!}$. *This represents the total number of ways to organize the participants into teams and assign captains accordingly.* ■

   Solution 2 : *Instead of arranging participants first and then grouping them, we can construct the teams step by step. Since each team requires a captain, we first select* k *captains from the* $4k$ *participants. The number of ways to do this is:* $\binom{4k}{k}$. *Af-*

ter selecting captains, the remaining $3k$ participants must be grouped into k teams of size 3. The number of ways to partition these participants into teams is: $\frac{(3k)!}{(3!)^k k!}$ where $(3k)!$ accounts for arranging the remaining participants, $(3!)^k$ corrects for the ordering within each group, and $k!$ removes overcounting due to indistinguishable teams. Since the captains are already chosen and distinct, they can be placed into teams in $k!$ ways. Therefore, the total number of ways to form the teams and assign captains is: $\binom{4k}{k} \cdot \frac{(3k)!}{(3!)^k k!} \cdot k! = \frac{(4k)!4^k}{(4!)^k k!}$. ■

10. Let a, b, c, d be four real numbers such that $a \geq b \geq c \geq d > 0$ and $a + b + c + d = 1$. Prove that $(a + 2b + 3c + 4d)a^a b^b c^c d^d < 1$.

Solution : *The* **weighted AM-GM inequality** *with weights a, b, c, d gives:*

$$a^a b^b c^c d^d \leq a \cdot a + b \cdot b + c \cdot c + d \cdot d = a^2 + b^2 + c^2 + d^2,$$

so it is enough to prove that $(a+2b+3c+4d)(a^2+b^2+c^2+d^2) < 1 = (a+b+c+d)^3$. This can be done in several ways, including the following:

$$(a+b+c+d)^3 > a^2(a+3b+3c+3d) + b^2(3a \cdot$$
$$+ d^2(3a+3b+3c+d) \geq (a^2$$

■

SECTION B: 10 **marks each**

11. Find all functions $f : \mathbb{R} \to \mathbb{R}$, $x \mapsto f(x)$ (defined on the real numbers and taking real values), which satisfy the equation $f(x)f(y) = f(x+y) + 4xy$ for all real numbers x and y.

Solution : *Setting $y = 0$ yields $f(x)f(0) = f(x)$ hence either $f(x) = 0$ for every $x \in \mathbb{R}$ or $f(0) = 1$. However, $f(x) = 0$ for all $x \in \mathbb{R}$ is not a solution since rhw functional equation reduces to $4xy = 0$ which implies that either x or y must be 0. Now substituting $y = -x$ yields $f(x)f(-x) = f(0) - 4x^2 = 1 - 4x^2$. In particular, if $x = \frac{1}{2}$, then $f\left(\frac{1}{2}\right) f\left(-\frac{1}{2}\right) = 1 - 4\left(\frac{1}{2}\right)^2 = 0$, hence either $f\left(\frac{1}{2}\right) = 0$ or $f\left(-\frac{1}{2}\right) = 0$. Now, if $f\left(\frac{1}{2}\right) = 0$, then (using $y = \frac{1}{2}$) we have $0 = f(x)f\left(\frac{1}{2}\right) = f\left(x + \frac{1}{2}\right) + 2x$ for every $x \in \mathbb{R}$, i.e., $f(x) = 1 - 2x$. On the other hand, if $f\left(-\frac{1}{2}\right) = 0$, then (using $y = -\frac{1}{2}$) we have $0 = f(x)f\left(-\frac{1}{2}\right) = f\left(x - \frac{1}{2}\right) - 2x$ for every $x \in \mathbb{R}$, i.e., $f(x) = 1 + 2x$. $f(-1) = 0$, then (using $y = -1$) we have $0 = f(x)f(-1) = f(x-1) - x$ for every $x \in \mathbb{R}$, i.e., $f(x) = 1 + x$. Therefore, $f(x) = 1 - 2x$ or $f(x) = 1 + 2x$ for all $x \in \mathbb{R}$.* ■

12. In an acute, non-isosceles triangle ABC, the midpoints of AC and AB are B_1 and C_1 respectively. A point D lies on BC with C between B and D. The point F is such that $\angle AFC$ is a right angle and $\angle DCF = \angle FCA$. The point G is such that $\angle AGB$ is a right angle and $\angle CBG = \angle GBA$. Prove that B_1, C_1, F and G are collinear.

Solution 1 :

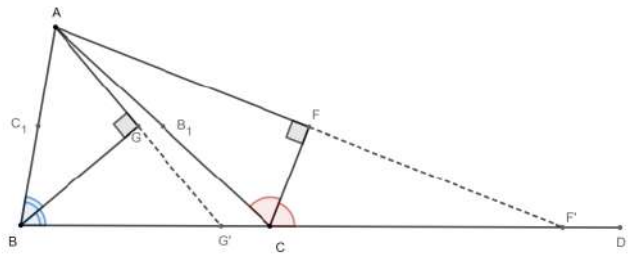


Figure 1:

Extend AF and AG to meet BC at F′ and G′ respectively. As a consequence of ASA, we have: $\angle FCA = \angle FCF'$, $CF = CF$, $\angle AFC = 90° = \angle CFF'$ implying that $\Delta AFC \cong \Delta FCF'$. Similarly, $\angle GBA = \angle GBG'$, $BG = BG$, $\angle AGB = 90° = \angle BGG'$ implying that $\Delta AGB \cong \Delta BGG'$. Therefore, $AF = FF'$ and $AG = GG'$ so F, G, B_1, C_1 are the midpoints of AF', AG', AC, AB respectively. This means: $\frac{AF}{AF'} = \frac{AG}{AG'} = \frac{AB_1}{AC} = \frac{AC_1}{AB} = \frac{1}{2}$. Therefore, if we consider an enlargement with scale factor $\frac{1}{2}$ at A, then the line passing

through B, G', C, F' maps to a line passing through C_1, G, B_1, F proving these four points are collinear. ■

Solution 2 : *Since B_1, C_1 are the midpoints of AC, AB respectively, we have B_1C_1 is parallel to BC so $\angle B_1C_1A = \angle B$. Also, $\angle AGB = 90°$, so C_1 is the centre of circle AGB and we get: $\angle GC_1A = 2\angle GBA = \angle GBA + \angle CBG = \angle B = \angle B_1C_1A$. Therefore, G lies on B_1C_1. Similarly, $\angle AFC = 90°$ so B_1 is the centre of the circle AFC giving: $\angle FB_1A = 2\angle FC'A = \angle FC'A + \angle DCF = \angle DCA = 180° - \angle C$. And using B_1C_1 is parallel to BC, we get: $\angle AB_1C_1 = \angle C$ implying that $\angle AB_1C_1 + \angle FB_1A = 180°$ so F also lies on B_1C_1.* ■

13. Let S be a subset of $\{1, 2, \ldots, 2024\}$ which consists of 771 distinct numbers. Show that there are two distinct elements a and b of S such that $a + b$ is divisible by 8.

Solution : *Let $T = \{1, 2, \ldots, 2024\}$, S be a subset of T with $|S| = 771$ and $A = \{0, 1, \ldots, 7\}$ be the set representatives of possible remainders when an integer in T is divided by 8. For any $(a, b) \in S^2 - \{(x, x) : x \in S\}$, we have $a + b \equiv 0 \pmod 8$ if and only if $a \equiv -b \pmod 8$. Therefore, the proper choice should be in such away that, it is equivalent to one of the pairs $\{(0, 0), (1, 7), (2, 6), (3, 5), (4, 4)\}$. Since $|T| = 2024$, $|S| = 771$ and there are 8 residue classes, it follows by the pigeonhole principle that, for each $k \in T$, there are at-most $\frac{2024}{8} = 256$ numbers that leave a remainder k. We would put 256 numbers in either hole 1 or 7. Similarly, we could put 256 in hole 2 or 6. And another 256 in hole 3 or 5. And we could put 1 in holes 0 and 4. This would give us $256 + 256 + 256 + 1 + 1 = 770$. This therefore is the maximum number before two of the numbers add to a multiple of 8. With 771, we would therefore violate this property. Therefore, at least one of the residue classes in pairs must contain 2 or more elements. Specifically: if the residue class 0 contains at least 2 elements, then $a + b \equiv 0 \pmod 8$ for those two elements. If the residue class 1 contains at least 1 element and the residue class 7 contains at least 1 element, then $a + b \equiv 0 \pmod 8$ for those two elements, and similarly for other pairs of residue classes. This concludes the proof that there must be two distinct elements a, b such that $a + b \equiv 0 \pmod 8$.* ■

14. It is well known that for each positive integer n, $\sum_{k=1}^{n} k^3 = \frac{n^2(n+1)^2}{4}$. Determine whether or not there is a positive integer m such that $\sum_{k=m+1}^{2m} k^3$ is a square.

Solution : *Now $\sum_{k=m+1}^{2m} k^3 = \sum_{k=1}^{2m} k^3 - \sum_{k=1}^{m} k^3 = \frac{(2m)^2(2m+1)^2}{4} - \frac{m^2(m+1)^2}{4} = \frac{m^2(3m+1)(5m+3)}{4}$.*

- *If m is even, then $m = 2s$ for some $s \in \mathbb{Z}_{\geqslant 1}$ as $m \geqslant 1$. From the above expression, we get $\sum_{k=m+1}^{2m} k^3 = \frac{m^2(3m+1)(5m+3)}{4} = s^2(6s+1)(10s+3) = s^2(60s^2 + 16s + 3) \equiv 3s^2 \pmod 4$. Now, since $n^2 \equiv 0, 1 \pmod 4$, it follows that $3s^2 \equiv 0, 3 \pmod 4$, hence we must have $s \in 2\mathbb{Z}_{\geqslant 1}$, say $s = 2t$ for some $t \in \mathbb{Z}_{\geqslant 1}$. We obtain $s^2(60s^2 + 16s + 3) = 4t^2(240t^2 + 32t + 3)$. Again, for the new expression to be a square, $240t^2 + 32t + 3$ must also be a square. However, $240t^2 + 32t + 3 \equiv 3 \pmod 4$ implying $240t^2 + 32t + 3$ is not a square, hence there is no even m such that $\sum_{k=m+1}^{2m} k^3$ is a square.*
- *If m is odd, then $m = 2s + 1$ for some $s \in \mathbb{Z}_{\geqslant 0}$ as $m \geqslant 1$. From the above ex-*

pression, we get $\sum_{k=m+1}^{2m} k^3 = \frac{m^2(3m+1)(5m+3)}{4} = (2s+1)^2(3s+2)(5s+3) \equiv s^2+2 \pmod 4$. *Now, since* $n^2 \equiv 0, 1 \pmod 4$, *it follows that* $s^2+2 \equiv 2, 3 \pmod 4$, *both of which are none squares modulo* 4. *Therefore, there is no odd* m *such that* $\sum_{k=m+1}^{2m} k^3$ *is a square.*

In conclusion, there is no positive integer m *such that* $\sum_{k=m+1}^{2m} k^3$ *is a perfect square.* ■

15. (a) Mrs. Webb gave an exam in a mathematics class of five students. She entered the results in a random order into a spreadsheet, which recalculated the class average after each result was entered. Mrs. Webb noticed that after each result was entered, the average was always an integer. The scores (listed in ascending order) were $71, 76, 80, 82$, and 91. What was the last result Mrs. Webb entered?

Solution : *Let* x_1, x_2, x_3, x_4 *and* x_5 *be the results to be entered in that order, the sequence of the averages is*

$$x_1, \frac{x_1+x_2}{2}, \frac{x_1+x_2+x_3}{3}, \frac{x_1+x_2+x_3+x_4}{4}, \frac{x_1+x_2, x_3+x_4+x_5}{5}$$

and are all integers. This implies

(i) x_1 *is an integer*

(ii) x_1+x_2 *is divisible by* 2.

(iii) $x_1+x_2+x_3$ *is divisible by* 3.

(iv) $x_1+x_2+x_3+x_4$ *is divisible by* 4.

(v) $x_1+x_2, x_3+x_4+x_5$ *is divisible by* 5.

Property (15(a)v) is satisfied since $x_1+x_2+x_3+x_4+x_5 = 400$ *is divisible by* 5 *other properties give* $x_1+x_2+x_3+x_4 \equiv 0 \pmod 4, x_1+x_2+x_3 \equiv 0 \pmod 3$ *and* $(71, 76, 80, 82, 91) \equiv (2, 1, 2, 1, 1) \mod 3$. *It follows that the only way to get a number divisible by* 3 *is by adding three of the ones and so* $76+82+91 \equiv 0 \pmod 3$, *so* $x_1+x_2+x_3 = 249 \equiv 0 \pmod 3$ *and now* x_4 *and* x_5 *are obtained for either* 71 *or* 80 *and for* $x_1+x_2+x_3+x_4 \equiv 0 \pmod 4$ *thus* $x_4 = 71$ *and therefore the last result* x_5 *entered was* 80, *that is* $x_5 = 80$. ■

(b) Suppose, $2a^2+b^2-4a-6b = -11$, where a and b are integers. What is the value of $(a^{ba})(b^a)$?

Solution : $2a^2+b^2-4a-6b = 2(a^2-2a)+b^2-6b = -11 \Rightarrow 2(a^2-2a+1)+(b^2-6b+9 = -11+2+9 = 0$ *thus* $2(a-1)^2+(b-3)^2 = 0$ *then* $a = 1$ *and* $b = 3$ *Now* $(a^{ba})(b^a) = 3$. ■

END